

PEF 6000 - Special topics of dynamics of structures
**Module 5 - Introduction to fluid-structure
interaction**
Presentation 1: Flow around bluff bodies

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Outline

- 1 Objectives
- 2 Flow around immersed bodies - introduction
- 3 Flow around a fixed cylinder

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- Examples of references: Textbooks by Blevins (2001), Païdoussis et al (2011), Naudascher & Rockwell (2005), the reviews by Sarpkaya (2004) and Williamson & Govardhan (2004), habilitation thesis Franzini (2019) and selected papers.

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Definitions

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- Full-slip condition: The component in the direction of the normal to the surface of the relative velocity is null.

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- Ideal fluids: Inviscid fluid $\rightarrow$ Full-slip condition (**basic hypothesis for the potential flow theory.**)

Navier-Stokes equations

Equation of motion for the fluid particles:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{\nabla p}{\rho} + \mathbf{g} + \nu \nabla^2 \mathbf{u} \quad (1)$$

Continuity equation for non-compressible flows:

$$\nabla \cdot \mathbf{u} = 0 \quad (2)$$

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- Far from the body (far from the boundary layer), the viscous effects are irrelevant and the solution from the potential theory leads to acceptable results;

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- The presence of corners defines the separation points;
- For a cylinder, the separation point is dependent on the Reynolds number (to be better discussed).

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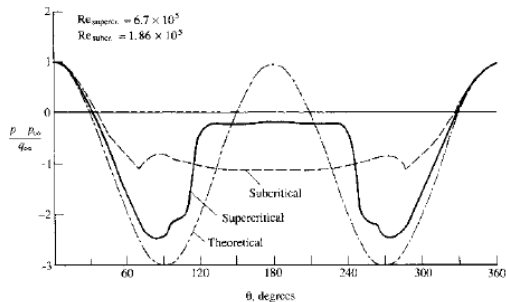
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- Focus of the class: Circular cylinders.

Bluff body

- Distribution of the pressure coefficient $C_P = \frac{p - p_\infty}{1/2 \rho U_\infty^2}$

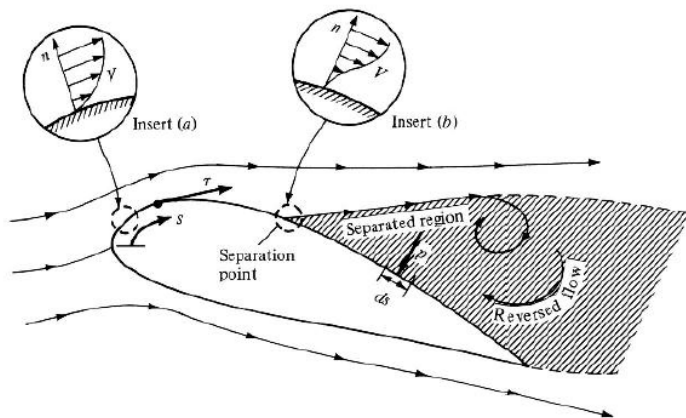
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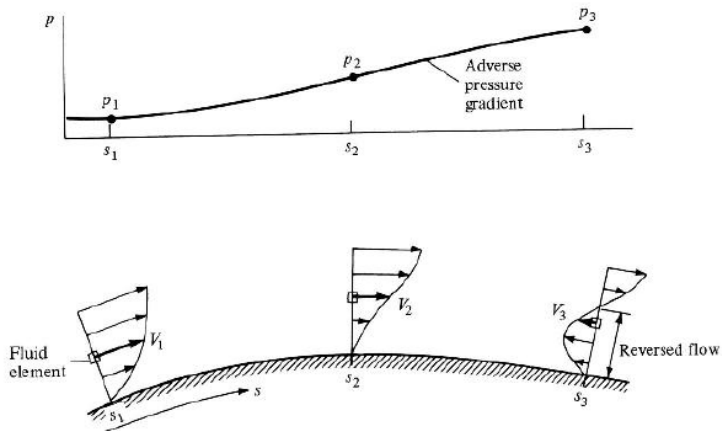
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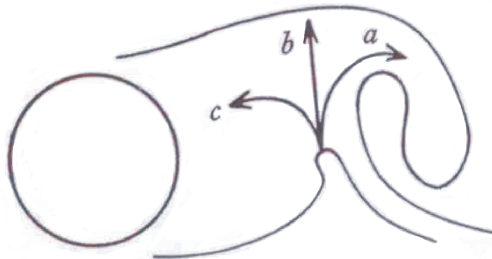
Fixed cylinder

- As already mentioned, the flow separation in problems involving circular cylinders strongly depends on the Reynolds number $Re = U_{\infty} D / \nu$ (occurring around 80° , measured from the frontal stagnation point);

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- As already mentioned, the flow separation in problems involving circular cylinders strongly depends on the Reynolds number $Re = U_{\infty} D / \nu$ (occurring around 80° , measured from the frontal stagnation point);
- The flow separation gives rise to two free shear layers of opposite circulation. The interaction of the free shear layers causes the formation of vortexes.

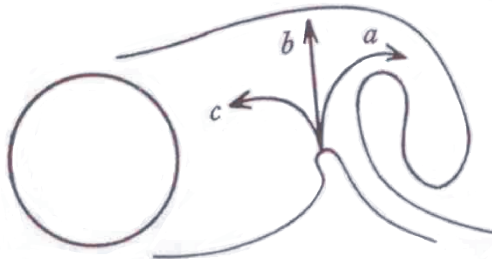
Bidimensional model proposed by Gerrard (1966)



Extracted from Gerrard (1966).

Gerrard stated that the fluid particles follow on of three distinct paths:

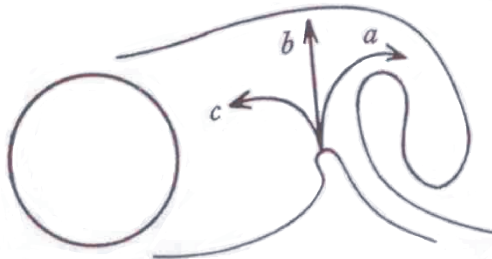
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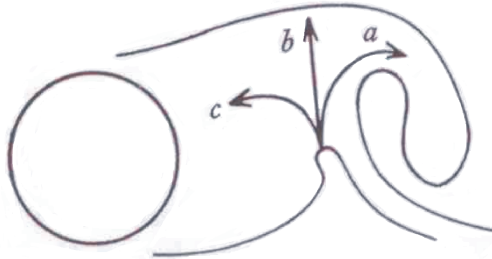


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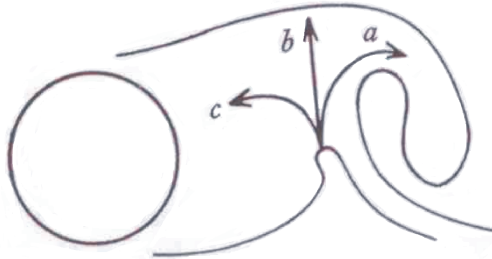


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- **Path c:** Associated with the new vortex, with opposite circulation.

Vortex shedding

- The described process periodically repeats, given rise to the vortex-shedding phenomenon
→ **vón Kármán wake**.

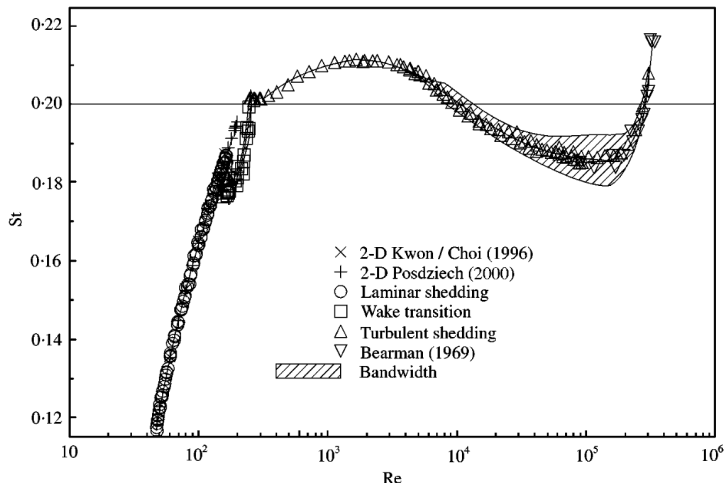
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- The vortex-shedding frequency is f_s and the Strouhal number is calculated as $St = f_s D / U_\infty$. For circular cylinders and for a broad range of Reynolds number, $St \approx 0.20$.
- Since the velocity field (not the incoming flow) is oscillatory, the pressure field also depends on time. This leads to oscillatory hydrodynamic forces acting on the cylinder.

Vortex shedding



Extracted from Norberg (2001).

Hydrodynamic forces

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- Forces associated with pressure, of great importance for bluff bodies;
- Forces associated with friction: Associated with the shear stress (important on blunt bodies).
- Forces associated with waves: Highly important in flows with the presence of free-surfaces (ships, for example).

Hydrodynamic forces in bidimensional bodies

We decompose the hydrodynamic forces into two terms:

- Drag force (F_D): Component of the force into the direction of the relative velocity. Drag coefficient $C_D = \frac{F_D}{1/2 \rho U_\infty^2 DL}$

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- In the flow around a bidimensional cylinder, the force coefficients can be well approximated by

$$C_L(t) = \hat{C}_L \sin(\omega_s t)$$

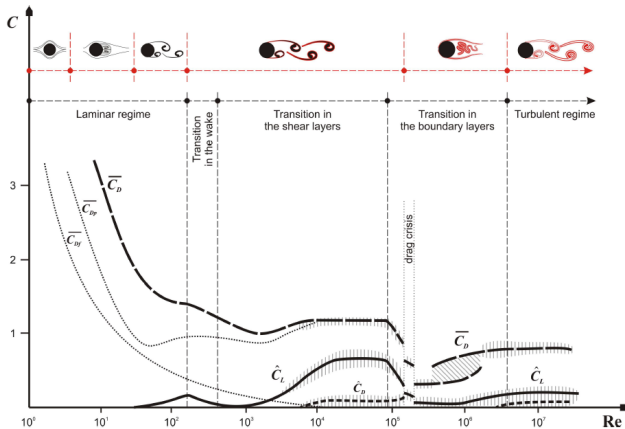
$$C_D(t) = \bar{C}_D + \hat{C}_D \sin(2\omega_s t + \phi)$$

Influence of the Reynolds number

Reynolds number range	Name of the regime	Characteristic feature	General properties
$Re < 1$		Creeping flow	Laminar regime
$3-5 < Re < 30-40$	L	Steady separation (recirculation bubble)	
$30-40 < Re < 150-300$	L	Periodic laminar shedding	
$150-300 < Re < 1 \times 10^5-2 \times 10^5$	TrW and TrSL		Subcritical regime: laminar separation transition in shear layer turbulent wake
$150-200 < Re < 200-250$	TrW1	Transition of laminar vortices in wake,	
$200-250 < Re < 350-500$	TrW2	Transition of irregular vortex during its formation	
$350-500 < Re < 1 \times 10^3-2 \times 10^3$	TrSL1	Development of transition waves in free shear layer	
$1 \times 10^3-2 \times 10^3 < Re < 2 \times 10^4-4 \times 10^4$	TrSL2	Formation of transition vortices in free shear layer	
$2 \times 10^4-4 \times 10^4 < Re < 1 \times 10^5-2 \times 10^5$	TrSL3	Fully turbulent shear layer	
$1 \times 10^5-2 \times 10^5 < Re < 3.5 \times 10^5-6 \times 10^6$	TrBL		Critical regime: laminar separation turbulent reattachment turbulent separation Turbulent wake
$1 \times 10^5-2 \times 10^5 < Re < 3 \times 10^5-3.1 \times 10^5$	TrS0/TrBL0	Onset of transition at separation point	
$3 \times 10^5-3.1 \times 10^5 < Re < 3.3 \times 10^5-3.4 \times 10^5$	TrS1/TrBL1	Single separation bubble regime	
$3.3 \times 10^5-3.4 \times 10^5 < Re < 3.6 \times 10^5-3.8 \times 10^5$		Unstable regime	
$3.6 \times 10^5-3.8 \times 10^5 < Re < 5 \times 10^5-1 \times 10^6$	TrS2/TrBL2	Two-bubble regime	
$5 \times 10^5-1 \times 10^6 < Re < 3.5 \times 10^6-6 \times 10^6$	TrS3/TrBL3	Supercritical regime—fragmented separation bubble.	
$3.5 \times 10^6-6 \times 10^6 < Re < 6 \times 10^6-8 \times 10^6$	TrBL4	Transcritical regime—partial transition	
$Re > 8 \times 10^6$	T	Postcritical regime—complete transition	

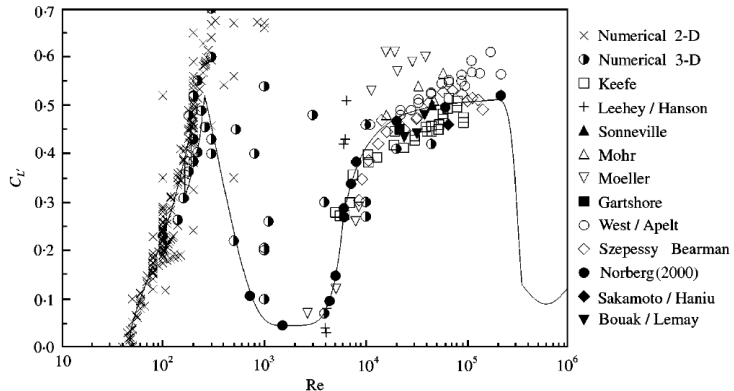
Extracted from Raghavan & Bernitsas (2011).

Influence of the Reynolds number



Extracted from Assi (2009).

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Extracted from Norberg (2001).

Technological application

- Consider a riser of external diameter equal to 8 in. Consider also a typical ocean current of intensity $U_\infty = 1$ m/s. Assuming $\nu = 10^{-6}$ m²/s, the Reynolds number is $Re = 1.01 \times 10^5$;
- For some floating units of cylindrical geometry, Reynolds number easily exceeds 10^7 (Fujarra et al. (2012));
- Proper characterization of the flow around cylinder at high Reynolds numbers is an open research topic of great relevance.